STUDY SHEET / MODULES 1-4

Amy's Cybersecurity Quick Review

Networking foundations. Read this first, then use the website for examples and detailed bilingual notes.

BEFORE YOU START

Use Packet Tracer to add devices, connect links, inspect logical and physical views, and practice on a small simulated network.

01 Network foundations & first setup

• A switch connects devices within a LAN; a router connects different networks.

• Devices need a correct IP address, subnet mask and default gateway to communicate as intended.

• Use the Cisco CLI to set names, passwords and management addresses. Save the configuration, then test with ping. A reliable network balances fault tolerance, scalability, quality of service and security.

02 How data moves

• Protocols are communication rules. The seven-layer OSI and four-layer TCP/IP models divide the work. Data is encapsulated when sent and de-encapsulated when received.

• An IP address identifies the destination across networks; a MAC address supports delivery on the current link. A switch learns source MAC addresses and forwards using destination MAC addresses.

• ARP maps a local IPv4 address to a MAC address. For a remote destination, resolve the gateway MAC. IPv6 uses Neighbor Discovery (ND).

• ping checks reachability; traceroute shows responding hops and helps locate connectivity problems.

REMEMBER THE TWO ADDRESSES

IP identifies the destination across networks. MAC gets a frame to the next device on the current Ethernet link. For a remote destination, the first frame goes to your gateway MAC.

Build it. Verify it.

03 IP addressing, subnets & routing

• IPv4 has 32 bits divided into network and host portions. The subnet mask identifies the boundary.

• Subnetting divides a larger network. VLSM uses different subnet sizes to match host needs; allocate the largest requirement first.

• Routers choose the most specific matching route. Static routes are configured manually; a default route handles traffic without a more specific match; floating static routes provide backup.

• Prefer encrypted SSH for device management. Troubleshoot interface state, addresses, gateways and routing tables.

04 VLANs & router configuration

• VLANs divide switching infrastructure into separate broadcast domains. Each VLAN normally corresponds to an IP subnet.

• An access port normally serves one data VLAN. A trunk carries multiple VLANs, distinguished with 802.1Q tags.

• Communication between VLANs needs a router or Layer 3 switch. Configure router interface addresses and enable interfaces with no shutdown. Hosts use their local router interface as the gateway.

Commands worth remembering

`show ip interface brief`	Inspect interface addresses and state
`show ip route`	Inspect the IPv4 routing table
`show vlan brief`	Inspect VLANs and access-port membership
`ping <address>`	Test ICMP reachability
`traceroute <address>`	Inspect responding hops (Windows: tracert)
`copy running-config startup-config`	Save the active configuration for restart

Troubleshooting order: link > interface state > IP/mask > VLAN > gateway > routes > end-to-end test. Check the return path, too.

Detailed notes: timememo.net/amy-cybersecurity/

Original study guide, reviewed October 27, 2026. Not an official Cisco publication. Use NetAcad for full lessons, videos, labs and assessments.